Demurrage & Detention Audit Engine – Production Spec v0.1

Working name: DwellCheck - October 6, 2026

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0. Scope and legal ground truth

Goal. Ingest bills of lading (B/L), carrier/MTO/NVOCC D&D invoices, carrier and terminal event data, and drayage GPS pings; compute what the customer *actually* owes; and auto-generate dispute packets that cite federal rules, filed inside every deadline.

Rules the engine encodes (verified against eCFR, 46 CFR Part 541, current as of Oct 5 2026):

Rule	Text (summary)	Engine effect
Sec. 541.5	Missing any required minimum info **eliminates the obligation to pay**	Field-completeness check = strongest, cheapest win
Sec. 541.6(a)	Must show B/L no., container no., port of discharge (imports), **basis for why billed party is liable**	R1-R4
Sec. 541.6(b)	Invoice date, due date, free days, free-time start/end, availability date (imports), earliest return date (exports), specific charged dates	R5-R12
Sec. 541.6(c)	Total, applicable tariff/service-contract rule, specific rate(s)	R13-R15
Sec. 541.6(d)	Dispute contact, URL/QR to public dispute-doc page, defined timeframes	R16-R18
Sec. 541.6(e)	Certification of FMC compliance and that billing party's performance didn't cause the charges	R19-R20
Sec. 541.7(a)	Invoice must issue **within 30 days** of the date the charge was last incurred, else **not payable**	R21
Sec. 541.7(b)	NVOCC pass-through: 30 days from the invoice it received	R22
Sec. 541.7(c)	NVOCC that is also billed can notify upstream; gets +30 days	Deadline extender
Sec. 541.7(d)	Re-bill to the correct party only within the original 30-day window	R23
Sec. 541.8(a)	Billed party gets **>=30 days from invoice issuance** to request mitigation/refund/waiver	Dispute deadline
Sec. 541.8(b)	Billing party must attempt to resolve within 30 days of request	Escalation timer
46 USC 41310	Charge Complaint to FMC: needs B/L numbers + invoices; **carrier bears burden** of proving reasonableness under 46 CFR 545.5; refunds + civil penalties	Escalation path (Tier 3)
46 CFR 545.5	Interpretive rule: D&D must serve the incentive principle (moving cargo); charges when cargo *can't* move (no appointments, terminal closed, container unavailable, no empty-return location) are likely unreasonable	Reasonableness rules R30-R39

Note. FMC finalized updated Charge Complaint procedures (46 CFR Part 502) on Sept 1 2026; re-read before shipping Tier 3. Rules engine must be versioned by effective date (a 2023 invoice is judged on pre-May-28-2024 rules).

Scope v1. US imports/exports, ocean carriers (VOCC), MTOs, NVOCCs. Out of scope v1: rail storage, chassis per-diem from IEPs (different regime; v2), non-US ports.

---

1. System architecture

```
SOURCES
  Email inbox (invoices) | Carrier portals (browser workers) | EDI 310/315/322/810
  DCSA T&T API | Terminal portals | TMS / drayage GPS | Service contracts & tariffs
        |
        v
```

```
INGESTION:  Doc AI parser  +  Event normalizer  +  GPS geofencer
        |
        v
CORE:  Container Timeline -> Clock Engine (free time, chargeable days)
         -> Rules Engine (R1-R40 findings) --> Human Review Queue
        |
        v
OUTPUT:  Dispute Builder (packet PDF, letter, JSON)
         -> Submission Workers (email, portal, FMC complaint draft)
```

Services

Service	Stack	Responsibility
`ingest-mail`	Gmail/Graph API watcher	Pull invoice PDFs/XLS from AP inboxes; dedupe by hash
`ingest-portal`	Browser worker pool (Playwright)	Log into carrier/terminal portals with customer creds from vault; download invoices, availability, gate events
`ingest-edi`	AS2/SFTP + X12 parser	315 (status), 322 (terminal ops), 310 (freight receipt), 810 (invoice)
`ingest-api`	DCSA Track & Trace v2 + carrier APIs	Equipment/transport events
`ingest-gps`	Webhooks from TMS/ELD (Samsara, Motive, project44, etc.)	Truck pings, matched to container by dispatch
`docai`	Layout model + LLM extraction w/ JSON schema; confidence per field	Invoice, B/L, delivery order, EIR parsing
`timeline`	Postgres + event sourcing	Canonical container timeline, source precedence
`clock`	Deterministic TS/Python lib	Free time, chargeable days, calendar rules
`rules`	Versioned rule packs (YAML + code)	Findings with evidence pointers
`packet`	Template engine -> PDF + JSON	Dispute letters, exhibits, FMC complaint draft
`submit`	Browser workers + SMTP + carrier APIs	File disputes, capture confirmation
`review`	Web app	Human-in-the-loop queue
`ledger`	Postgres	Money: disputed, credited, refunded, fee earned

---

2. Canonical data schemas (JSON Schema 2020-12, abridged to fields that matter)

2.1 Shipment / B/L

```json
{
  "$id": "https://dwellcheck.io/schema/shipment.json",
  "type": "object",
  "required": ["shipment_id", "bl_number", "direction", "carrier_scac", "containers"],
  "properties": {
    "shipment_id": {"type": "string", "format": "uuid"},
    "tenant_id": {"type": "string"},
    "bl_number": {"type": "string", "pattern": "^[A-Z0-9]{6,20}$"},
    "house_bl_number": {"type": ["string", "null"]},
    "bl_type": {"enum": ["MASTER", "HOUSE", "SEAWAY"]},
    "direction": {"enum": ["IMPORT", "EXPORT"]},
    "carrier_scac": {"type": "string", "minLength": 2, "maxLength": 4},
    "nvocc_scac": {"type": ["string", "null"]},
    "service_contract_no": {"type": ["string", "null"]},
    "parties": {
```

```
      "type": "object",
      "properties": {
        "shipper": {"$ref": "#/$defs/party"},
        "consignee": {"$ref": "#/$defs/party"},
        "notify": {"$ref": "#/$defs/party"},
        "billed_party_of_record": {"$ref": "#/$defs/party"}
      }
    },
    "pol_unlocode": {"type": "string", "pattern": "^[A-Z]{5}$"},
    "pod_unlocode": {"type": "string", "pattern": "^[A-Z]{5}$"},
    "discharge_terminal_smdg": {"type": ["string", "null"]},
    "containers": {"type": "array", "items": {"$ref": "container.json"}},
    "documents": {"type": "array", "items": {"$ref": "#/$defs/docref"}}
  },
  "$defs": {
    "party": {"type": "object", "properties": {
      "name": {"type": "string"}, "scac_or_id": {"type": ["string", "null"]},
      "role": {"enum": ["SHIPPER", "CONSIGNEE", "NOTIFY", "BCO", "NVOCC", "FORWARDER", "TRUCKER"]}}},
    "docref": {"type": "object", "required": ["doc_id", "kind", "sha256"], "properties": {
      "doc_id": {"type": "string"}, "kind": {"enum": ["BL", "INVOICE", "DO", "EIR", "ARRIVAL_NOTICE", "TARIFF",
      ↪  "SC_AMENDMENT", "EMAIL", "SCREENSHOT"]},
      "sha256": {"type": "string"}, "source": {"type": "string"}, "captured_at": {"type": "string", "format": "date-time"}}}
  }
}
```

2.2 Container timeline event (canonical)

Mapped from DCSA event codes where possible.

```
{
  "$id": "https://dwellcheck.io/schema/event.json",
  "type": "object",
  "required": ["event_id", "container_no", "type", "occurred_at", "source", "confidence"],
  "properties": {
    "event_id": {"type": "string"},
    "container_no": {"type": "string", "pattern": "^[A-Z]{4}[0-9]{7}$"},
    "type": {"enum": [
      "VESSEL_ARRIVED", "DISCHARGED", "AVAILABLE_FOR_PICKUP", "HOLD_PLACED", "HOLD_RELEASED",
      "GATE_OUT_FULL", "DELIVERED_TO_CONSIGNEE", "EMPTY_AVAILABLE", "GATE_IN_EMPTY",
      "EMPTY_RETURN_REFUSED", "APPT_UNAVAILABLE", "TERMINAL_CLOSED",
      "EXPORT_EARLIEST_RETURN", "GATE_IN_FULL", "LOADED", "CUTOFF_CHANGED", "ROLLED"
    ]},
    "occurred_at": {"type": "string", "format": "date-time"},
    "local_tz": {"type": "string"},
    "location": {"type": "object", "properties": {
      "unlocode": {"type": "string"}, "facility_smdg": {"type": ["string", "null"]},
      "lat": {"type": ["number", "null"]}, "lng": {"type": ["number", "null"]}}},
    "hold_type": {"enum": [null, "CUSTOMS", "USDA", "FREIGHT", "LINE", "TERMINAL", "CTF", "OTHER"]},
    "source": {"enum": ["DCSA_API", "EDI_315", "EDI_322", "TERMINAL_PORTAL", "CARRIER_PORTAL", "GPS_GEOFENCE", "EIR",
    ↪  "INVOICE_STATED", "USER"]},
    "source_ref": {"type": "string", "description": "doc_id, API request id, or GPS ping id"},
    "confidence": {"type": "number", "minimum": 0, "maximum": 1},
    "superseded_by": {"type": ["string", "null"]}
  }
}
```

Source precedence when events conflict (per event type, highest wins unless confidence < 0.6): `EIR` > `TERMINAL_PORTAL` > `EDI_322` > `DCSA_API` > `EDI_315` > `GPS_GEOFENCE` > `CARRIER_PORTAL` > `INVOICE_STATED` > `USER`. Conflicts > 4 hours on a chargeable boundary -> finding `F_TIMELINE_CONFLICT` -> human review.

2.3 Invoice (parsed)

```
{
  "$id": "https://dwellcheck.io/schema/invoice.json",
  "type": "object",
  "required": ["invoice_id", "billing_party", "invoice_no", "lines", "fields"],
  "properties": {
    "invoice_id": {"type": "string"},
    "billing_party": {"type": "object", "properties": {
      "name": {"type": "string"}, "scac": {"type": ["string", "null"]},
      "kind": {"enum": ["VOCC", "MTO", "NVOCC"]}}},
    "billed_party": {"type": "object"},
    "invoice_no": {"type": "string"},
    "currency": {"type": "string", "default": "USD"},
```

```
    "fields": {
      "description": "Every Sec. 541.6 element, each with value + where found + extraction confidence",
      "type": "object",
      "properties": {
        "bl_numbers":           {"$ref": "#/$defs/f"},
        "container_numbers":    {"$ref": "#/$defs/f"},
        "port_of_discharge":    {"$ref": "#/$defs/f"},
        "liability_basis":      {"$ref": "#/$defs/f"},
        "invoice_date":         {"$ref": "#/$defs/f"},
        "due_date":             {"$ref": "#/$defs/f"},
        "free_days":            {"$ref": "#/$defs/f"},
        "free_time_start":      {"$ref": "#/$defs/f"},
        "free_time_end":        {"$ref": "#/$defs/f"},
        "availability_date":    {"$ref": "#/$defs/f"},
        "earliest_return_date": {"$ref": "#/$defs/f"},
        "charged_dates":        {"$ref": "#/$defs/f"},
        "total_amount":         {"$ref": "#/$defs/f"},
        "tariff_rule_ref":      {"$ref": "#/$defs/f"},
        "rates":                {"$ref": "#/$defs/f"},
        "dispute_contact":      {"$ref": "#/$defs/f"},
        "dispute_url_or_qr":    {"$ref": "#/$defs/f"},
        "dispute_timeframes":   {"$ref": "#/$defs/f"},
        "cert_fmc_compliance":  {"$ref": "#/$defs/f"},
        "cert_no_fault":        {"$ref": "#/$defs/f"}
      }
    },
    "lines": {"type": "array", "items": {"type": "object", "properties": {
      "container_no": {"type": "string"},
      "charge_type": {"enum": ["DEMURRAGE", "DETENTION", "COMBINED", "STORAGE", "PER_DIEM"]},
      "from_date": {"type": "string", "format": "date"},
      "to_date": {"type": "string", "format": "date"},
      "days": {"type": "integer"},
      "tiers": {"type": "array", "items": {"type": "object", "properties": {
        "days": {"type": "integer"}, "rate": {"type": "number"}, "amount": {"type": "number"}}}},
      "amount": {"type": "number"}}}},
    "issued_via": {"enum": ["EMAIL", "PORTAL", "EDI_810", "MAIL"]},
    "first_seen_at": {"type": "string", "format": "date-time"},
    "paid": {"type": "object", "properties": {
      "status": {"enum": ["UNPAID", "PAID", "PARTIAL", "PAID_UNDER_PROTEST"]},
      "amount": {"type": "number"}, "date": {"type": ["string", "null"]}}}
  },
  "$defs": {"f": {"type": "object", "properties": {
    "present": {"type": "boolean"},
    "value": {},
    "page": {"type": ["integer", "null"]},
    "bbox": {"type": ["array", "null"], "items": {"type": "number"}},
    "via_url": {"type": ["string", "null"], "description": "Element provided only via linked page"},
    "confidence": {"type": "number"}}}}
}
```

2.4 Claim (dispute) state

```
{
  "$id": "https://dwellcheck.io/schema/claim.json",
  "type": "object",
  "required": ["claim_id", "invoice_id", "state", "findings", "deadlines"],
  "properties": {
    "claim_id": {"type": "string"},
    "invoice_id": {"type": "string"},
    "rulepack_version": {"type": "string", "example": "fmc541-2024.05.28+545.5-2022"},
    "state": {"enum": [
      "DETECTED", "ANALYZED", "NO_ACTION", "NEEDS_EVIDENCE", "NEEDS_REVIEW", "APPROVED",
      "SUBMITTED", "ACKNOWLEDGED", "INFO_REQUESTED", "PARTIAL_CREDIT", "CREDITED",
      "DENIED", "ESCALATION_REVIEW", "FMC_COMPLAINT_FILED", "REFUNDED", "CLOSED_WON",
      "CLOSED_LOST", "EXPIRED"
    ]},
    "findings": {"type": "array", "items": {"$ref": "#/$defs/finding"}},
    "amounts": {"type": "object", "properties": {
      "invoiced": {"type": "number"}, "engine_owed": {"type": "number"},
      "disputed": {"type": "number"}, "credited": {"type": "number"},
      "refunded": {"type": "number"}, "fee_earned": {"type": "number"}}},
    "deadlines": {"type": "object", "properties": {
      "dispute_due": {"type": "string", "format": "date", "description": "invoice_date + max(30, stated window) days"},
      "carrier_response_due": {"type": ["string", "null"], "format": "date"},
      "internal_submit_by": {"type": "string", "format": "date", "description": "dispute_due - 5 business days"}},
```

```
    "submissions": {"type": "array", "items": {"type": "object", "properties": {
      "channel": {"enum": ["EMAIL", "CARRIER_PORTAL", "API", "FMC_CHARGE_COMPLAINT", "FMC_SMALL_CLAIM"]},
      "submitted_at": {"type": "string", "format": "date-time"},
      "confirmation_ref": {"type": ["string", "null"]},
      "packet_doc_id": {"type": "string"},
      "screenshot_doc_id": {"type": ["string", "null"]}}}},
    "reviewer": {"type": ["string", "null"]},
    "audit_log": {"type": "array", "items": {"type": "object"}}
  },
  "$defs": {"finding": {"type": "object", "required": ["rule_id", "severity", "amount_at_issue", "evidence"], "properties":
  ↪  {
    "rule_id": {"type": "string"},
    "category": {"enum": ["CONTENT_541_6", "TIMING_541_7", "MATH", "FREE_TIME", "REASONABLENESS_545_5", "PARTY",
    ↪  "DUPLICATE"]},
    "severity": {"enum": ["VOIDS_INVOICE", "VOIDS_LINE", "REDUCES", "INFO"]},
    "amount_at_issue": {"type": "number"},
    "explanation": {"type": "string"},
    "citation": {"type": "string"},
    "evidence": {"type": "array", "items": {"type": "string", "description": "doc_id#page or event_id"}},
    "confidence": {"type": "number"}}}}
}
```

State machine (allowed transitions)

```
DETECTED -> ANALYZED -> {NO_ACTION | NEEDS_EVIDENCE | NEEDS_REVIEW | APPROVED}
NEEDS_EVIDENCE -> ANALYZED              (new docs/events arrive)
NEEDS_REVIEW -> {APPROVED | NO_ACTION}
APPROVED -> SUBMITTED -> ACKNOWLEDGED -> {INFO_REQUESTED | PARTIAL_CREDIT | CREDITED | DENIED}
INFO_REQUESTED -> SUBMITTED
PARTIAL_CREDIT | DENIED -> ESCALATION_REVIEW -> {FMC_COMPLAINT_FILED | CLOSED_LOST}
FMC_COMPLAINT_FILED -> {REFUNDED | CLOSED_LOST}
CREDITED | REFUNDED -> CLOSED_WON
any pre-SUBMITTED state past deadlines.dispute_due -> EXPIRED (alert, still allow Tier 3 if rule allows)
```

---

3. Clock engine (deterministic)

```python
def chargeable_days(direction, timeline, tariff, sc):
    # 1. Resolve free-time start
    if direction == "IMPORT":
        start = max(timeline.first("DISCHARGED"),
                    timeline.first("AVAILABLE_FOR_PICKUP"))          # availability, not discharge
        dem_end  = timeline.first("GATE_OUT_FULL")
        det_start= dem_end
        det_end  = timeline.first("GATE_IN_EMPTY")
    else:
        start    = timeline.first("EMPTY_PICKUP")                    # detention out
        dem_start= max(timeline.first("GATE_IN_FULL"),
                       timeline.first("EXPORT_EARLIEST_RETURN"))
        dem_end  = timeline.first("LOADED")

    # 2. Free days from service contract first, else tariff
    free = sc.free_days(charge_type) or tariff.free_days(charge_type)
    calendar = sc.calendar or tariff.calendar         # CALENDAR | WORKING (excl. Sat/Sun/holidays)

    # 3. Days the clock must stop (545.5 incentive principle + contract terms)
    stops = union(
        intervals(timeline, "HOLD_PLACED", "HOLD_RELEASED", where=hold_not_caused_by_customer),
        intervals(timeline, "TERMINAL_CLOSED"),
        days_with(timeline, "APPT_UNAVAILABLE", min_attempts=1),
        days_with(timeline, "EMPTY_RETURN_REFUSED"),
        sc.excluded_days, tariff.excluded_days
    )

    # 4. Walk days in port-local tz; count charged days after free time exhausted
    ...
    return {"free_start", "free_end", "charged_dates": [...], "tiers": [...], "amount": ...}
```

Rules: all dates evaluated in **port-local timezone**; a partial day counts as a full day only if tariff says so; tier boundaries follow tariff (e.g., days 1-4 @ \$X, 5+ @ \$Y); free time never starts before availability for imports.

4. Rules engine – the 40 rules

Each rule: `id`, `applies_if`, `logic`, `severity`, `amount`, `citation`, `evidence_needed`, `auto_threshold` (min confidence to skip human review).

4.1 Content rules (Sec. 541.5 / Sec. 541.6) – "void the invoice"

ID	Check	Severity	Auto >=
R1	B/L number(s) present	VOIDS_INVOICE	0.95
R2	Container number(s) present	VOIDS_INVOICE	0.95
R3	Port of discharge present (imports)	VOIDS_INVOICE	0.95
R4	Basis for billed party's liability stated	VOIDS_INVOICE	0.90
R5	Invoice date	VOIDS_INVOICE	0.95
R6	Due date	VOIDS_INVOICE	0.95
R7	Free days stated	VOIDS_INVOICE	0.95
R8	Free-time start date	VOIDS_INVOICE	0.95
R9	Free-time end date	VOIDS_INVOICE	0.95
R10	Container availability date (imports)	VOIDS_INVOICE	0.95
R11	Earliest return date (exports)	VOIDS_INVOICE	0.95
R12	Specific charged dates	VOIDS_INVOICE	0.95
R13	Total amount	VOIDS_INVOICE	0.98
R14	Applicable tariff/SC rule reference	VOIDS_INVOICE	0.90
R15	Specific rate(s)	VOIDS_INVOICE	0.95
R16	Dispute contact	VOIDS_INVOICE	0.90
R17	URL/QR to public dispute-doc page (and page resolves)	VOIDS_INVOICE	0.90
R18	Defined dispute + resolution timeframes, >=30 days to dispute	VOIDS_INVOICE	0.90
R19	Certification: consistent with FMC rules incl. 545.5	VOIDS_INVOICE	0.90
R20	Certification: billing party's performance didn't cause charges	VOIDS_INVOICE	0.90

```yaml
- id: R17
  applies_if: invoice.issue_date >= "2024-05-28"
  logic: |
    f = invoice.fields.dispute_url_or_qr
    if not f.present: FAIL
    resp = worker.fetch(f.value, follow_redirects=true, anonymous=true)
    if resp.status != 200 or resp.requires_login: FAIL("not publicly accessible")
    if not llm.classify(resp.text, "describes documentation needed for mitigation/refund/waiver"): FAIL
  citation: "46 CFR 541.6(d)(2); 541.5"
  evidence_needed: [invoice_pdf, url_fetch_screenshot]
```

Accuracy guard: element provided only on a linked page counts as present only if the rule text allows it (R17 explicitly digital; others must be on/with the invoice). Low-confidence "absent" (<0.9) -> human review; false "missing" claims damage credibility with carriers.

4.2 Timing rules (Sec. 541.7)

ID	Logic	Severity
R21	`invoice_date - last_charged_date > 30 days` (VOCC/MTO)	VOIDS_INVOICE
R22	NVOCC: `nvocc_invoice_date - upstream_invoice_date > 30`	VOIDS_INVOICE
R23	Re-bill to corrected party after original 30-day window	VOIDS_INVOICE
R24	Stated dispute window < 30 days from issuance	VOIDS_INVOICE (via R18) + note

`invoice_date` source: stated date, cross-checked against email received timestamp / portal posting date; if stated date precedes first availability to customer by > 3 days, flag `F_BACKDATED` -> review.

4.3 Math & free-time rules

ID	Logic	Severity
R25	sum line amounts != total (+-$1)	REDUCES
R26	Rate != SC/tariff rate for that tier & date	REDUCES (difference)
R27	Free days < contract free days	REDUCES
R28	Free time started before availability (imports)	REDUCES
R29	Charged dates outside engine's chargeable window	REDUCES
R40	Duplicate: same container x charge type x overlapping dates on another invoice	VOIDS_LINE

4.4 Reasonableness rules (46 CFR 545.5 incentive principle)

ID	Evidence	Severity
R30	Terminal closed / no gate hours on charged day	REDUCES (those days)
R31	No pickup appointments available (portal screenshots, >=1 attempt/day logged)	REDUCES
R32	Container not available (hold not caused by customer, e.g. line or terminal hold; rolled/stacked unworkable)	REDUCES
R33	Empty return refused / no return location designated	REDUCES
R34	Customer gave notice and carrier/MTO failed to act (e.g., dual-transaction rule, chassis shortage notice)	REDUCES
R35	Charges continued during government closure, strike, weather emergency at port	REDUCES
R36	Export: earliest return date changed after booking; charged before new ERD	REDUCES
R37	Export: vessel delay/roll moved cutoff; demurrage charged for carrier's delay	REDUCES

ID	Evidence	Severity
R38	Billed party is not the party with contractual relationship / control (e.g., trucker or wrong consignee)	VOIDS_INVOICE
R39	GPS shows truck at terminal gate in queue during window but turned away (geofence dwell >= 45 min, no GATE_OUT)	REDUCES (supporting R31)

```yaml
- id: R39
  applies_if: gps_available and charge_type in [DEMURRAGE, COMBINED]
  logic: |
    for day in charged_dates:
      visits = gps.geofence_visits(terminal_polygon, day, truck in dispatch_for(container))
      if any(v.dwell_min >= 45 and not timeline.has("GATE_OUT_FULL", day) for v in visits):
        mark day as "attempted, denied"
  evidence: [gps_track_png, geofence_polygon_geojson, dispatch_record]
  citation: "46 CFR 545.5(c)(2) incentive principle; carrier bears burden under 46 U.S.C. 41310(b)(2)"
  auto_threshold: never     # reasonableness findings always human-reviewed in v1
```

---

5. Automation workflows

W1 – Ingest & match (continuous)

1. `ingest-mail` sees new message to `ap@` / `dnd@` -> classify (invoice / arrival notice / DO / other).
2. `docai` extracts invoice -> `invoice.json` with per-field confidence + bbox.
3. Match to shipment by B/L, then container; if no shipment, create stub and request B/L from TMS or portal worker.
4. Portal worker pulls missing events (availability, gate in/out, holds) for each container from terminal portal -> `event.json`.
5. GPS service pulls pings for dispatched trucks +-2 days around charged window.
6. Emit `ANALYZE(invoice_id)`.

W2 – Analyze

1. Build timeline; resolve conflicts by precedence.
2. Clock engine -> `engine_owed`.
3. Run rules (pack chosen by invoice date). Aggregate: if any `VOIDS_INVOICE` with confidence >= threshold -> `disputed = invoiced`; else `disputed = invoiced - engine_owed`.
4. Thresholds: `disputed < $150` and no VOID finding -> `NO_ACTION` (batch in monthly summary letter). Else `NEEDS_REVIEW` or `APPROVED` per auto rules (Sec. 7).
5. Compute deadlines; schedule reminders at D-10, D-5, D-2.

W3 – Build packet

Contents (PDF, bookmarked) + machine JSON: 1. Cover letter (template T1) with summary table of findings and amount. 2. Exhibit A: invoice with highlighted missing/incorrect fields (bbox overlays). 3. Exhibit B: container timeline table + source of each event. 4. Exhibit C: tariff/SC excerpt with applicable rule. 5. Exhibit D: evidence (terminal screenshots, appointment-unavailable logs, GPS map, EIRs). 6. Exhibit E: rule citations (46 CFR 541.5-541.8, 545.5 text).

W4 – Submit

Channel priority: carrier dispute portal/API -> dispute email named on invoice (Sec. 541.6(d)(1)) -> certified letter. Browser worker steps for portal submission (generic):

```
login(vault[tenant][carrier]) -> nav "Disputes"/"Invoice Dispute" -> select invoice_no
-> reason code map (Sec. 5.2) -> paste T2 summary (<= 2,000 chars) -> upload packet.pdf
-> submit -> capture confirmation id + full-page screenshot -> claim.submissions.push(...)
-> state SUBMITTED
on failure: retry x2 with fresh session; then fallback to email channel; never miss internal_submit_by
```

W5 – Track & respond

- Poll portal / parse replies daily. LLM classifies reply: `ACK | CREDIT_FULL | CREDIT_PARTIAL | DENIED | INFO_REQUEST`.
- Carrier must attempt resolution in 30 days (Sec. 541.8(b)); at day 31 with no resolution -> auto follow-up T3, flag for escalation review.
- Credits reconciled against next carrier statement; fee invoiced to customer only on **realized** credit/refund.

W6 – Escalate (Tier 3)

For denied/partial claims with VOID findings or strong 545.5 evidence and amount >= $2,500 (configurable): prepare **FMC Charge Complaint** draft (B/L numbers, invoices, alleged violations of 46 USC 41104(a)/41102, evidence). Customer signs and submits as the complainant (any person may submit under Sec. 41310). Option to route to partner maritime counsel for formal/small-claims proceedings.

5.1 Mock payloads

Dispute API / portal JSON (normalized internal -> carrier adapter)


```
{
  "dispute_ref": "DWC-2026-000418",
  "billing_party_scac": "MAEU",
  "invoice_no": "DD-7754120",
  "invoice_date": "2026-08-14",
  "bl_numbers": ["MAEU244118812"],
  "containers": [
    {"container_no": "MSKU7731042", "charge_type": "DEMURRAGE",
     "invoiced_amount": 3150.00, "disputed_amount": 3150.00,
     "reason_codes": ["MISSING_REQUIRED_INFO", "LATE_INVOICE"],
     "findings": [
       {"rule": "R10", "citation": "46 CFR 541.6(b)(6)", "text": "Invoice omits container availability date."},
       {"rule": "R21", "citation": "46 CFR 541.7(a)", "text": "Last charge incurred 2026-07-09; invoice issued 2026-08-14
       ↪  (36 days)."}
     ]}
  ],
  "requested_action": "WAIVER",
  "contact": {"name": "AP Team, Acme Imports", "email": "dnd@acme.example", "phone": "+1-410-555-0100"},
  "attachments": [{"name": "DWC-2026-000418-packet.pdf", "sha256": "..."}]
}
```


EDI 315 event -> canonical


```
{"src": {"B4_03": "OA", "B4_07": "MSKU", "B4_08": "7731042", "DTM": "20260702", "R4_LOC": "USBAL"},
 "out": {"container_no": "MSKU7731042", "type": "GATE_OUT_FULL", "occurred_at": "2026-07-02T14:12:00-04:00",
         "location": {"unlocode": "USBAL"}, "source": "EDI_315", "confidence": 0.9}}
```


GPS geofence visit


```
{"truck_id": "T-114", "container_no": "MSKU7731042", "facility": "Seagirt Marine Terminal",
 "enter": "2026-06-30T07:41:00-04:00", "exit": "2026-06-30T10:02:00-04:00", "dwell_min": 141,
 "gate_out_in_window": false, "ping_count": 47, "polygon_version": "smt-2026-03"}
```


5.2 Reason-code mapping (internal -> typical carrier portal picklists)

Internal	Typical picklist
MISSING_REQUIRED_INFO	“Invoice incomplete / regulatory”
LATE_INVOICE	“Invoice issued late”
FREE_TIME_ERROR	“Free time incorrect”
RATE_ERROR	“Tariff/contract rate incorrect”
CONTAINER_UNAVAILABLE	“Container not available / hold”
NO_APPOINTMENTS	“Terminal congestion / no appointments”
EMPTY_RETURN_REFUSED	“Empty return not accepted”
WRONG_PARTY	“Not liable party”
DUPLICATE	“Duplicate billing”

5.3 Letter templates

T1 – Dispute cover letter (summary) > RE: Request for waiver – Invoice {invoice_no}, B/L {bl}, Container(s) {containers} > Under 46 CFR 541.8(a), {customer} requests waiver of {disputed_amount}. The invoice {omits required element(s): list} (46 CFR 541.6), which under 46 CFR 541.5 eliminates any obligation to pay. {Additionally/Alternatively: timing and reasonableness findings.} Please resolve within 30 days per 46 CFR 541.8(b). Exhibits attached.

T3 – 30-day follow-up: notes no resolution within Sec. 541.8(b) window; requests written decision within 10 business days; reserves rights under 46 USC 41310.

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6. Error recovery

Failure	Detection	Recovery
OCR/extraction low confidence	field conf < threshold	Re-run with alternate model; then human field-verify UI (click the field on the PDF)
Missing B/L or events	timeline gaps on chargeable boundary	Portal worker fetch; ask customer's TMS; `NEEDS_EVIDENCE` with checklist; never assume
Portal login fails	401/MFA/captcha	Vault re-auth flow -> customer prompt; switch to email channel if deadline < 3 days
Portal layout changed	selector/step failure x2	Mark adapter `DEGRADED`, fallback channel, page on-call to update playbook
Event conflicts	precedence tie or >4h discrepancy	`F_TIMELINE_CONFLICT` -> review with both sources shown
GPS gaps	> 20 min ping gap in window	Don't use R39 for that day
Deadline risk	`internal_submit_by` reached, not SUBMITTED	Escalate to reviewer on-call; submit minimal content-rule-only dispute to preserve rights
Duplicate submission	idempotency key `(tenant, invoice_no, version)`	Block second submit; append supplement instead
Carrier re-issues corrected invoice	new invoice same containers/dates	Link as `supersedes`; re-run R21/R23 (correction outside 30 days is still late)
Rule pack change	new FMC rule / effective date	Versioned packs; re-analyze open claims; never re-judge closed ones
Payment already made	invoice.paid = PAID	Request refund not waiver; Tier 3 eligible ("refund of charges paid")

All worker actions are idempotent and logged with screenshots; every claim is rebuildable from its event log.

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7. Human-in-the-loop boundaries

Auto-submit allowed (no human) only when ALL hold: - Findings are content (R1-R20) or timing (R21-R23) rules only. - Every finding confidence >= its auto threshold, and invoice field extraction verified by two independent passes. - Disputed amount <= $5,000 per invoice (tenant-configurable down, never up past $25k). - Customer enabled auto-submit for that billing party. - Not the first dispute against that billing party for this tenant (first one always reviewed).

Always human-reviewed: - Any reasonableness finding (R30-R39) or wrong-party finding (R38). - Anything sent to the FMC, any legal threat language, any formal/small claim. - Statements of fact made on the customer's behalf beyond documents (e.g., "we attempted pickup"). - Relationship-sensitive carriers flagged by the customer (key accounts). - Disputes > $5,000 or bundle > $25,000. - Disagreement between engine and invoice stated dates > 1 day.

Customer-only actions (product never does): - Signing or verifying FMC complaints (complainant is the customer; representation before FMC limited to admitted practitioners – partner counsel for formal cases). - Paying invoices or accepting settlements below 50% of disputed amount. - Waiving rights or signing releases.

Reviewer UI: side-by-side invoice PDF (highlights), timeline, clock math, findings with toggle, edited letter preview; actions Approve / Edit / Request evidence / Drop; target SLA 4 business hours; reviewer decisions train threshold calibration.

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8. Security, data, and metrics

- Customer portal credentials in a vault (per-tenant KMS keys); workers get short-lived decrypt; no credential reaches LLM context.
- SOC 2 Type II path from month 1; data retention 7 years (tax/audit) configurable.
- PII minimal; GPS stored only for dispatched trucks within +-2 days of charged windows.

North-star metrics: $ recovered / $ invoiced; win rate by rule and billing party; median days to credit; % auto-submitted; false-finding rate (target $< 2\%$); deadline misses (target 0).

Business model hook: 25-35% of realized credits/refunds; free 12-month look-back audit (only invoices still within dispute windows are disputable – older ones feed Tier 3 if paid and violating).

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9. MVP build order (8 weeks)

1. Wk 1-2: invoice parser + Sec. 541.6 completeness checker (R1-R20) + R21 late-invoice. Backtest on 500 real invoices from 3 design partners. This alone is a product.
2. Wk 3-4: timeline + clock engine from terminal portals/EDI; R25-R29; packet builder; email submission.
3. Wk 5-6: review UI, deadline scheduler, reply classifier, ledger and fee billing.
4. Wk 7-8: portal adapters for top 5 carriers by tenant volume; R30-R33 with screenshots; GPS R39 for tenants with ELD.
5. Post-MVP: FMC Charge Complaint drafting, NVOCC flows, chassis per-diem (IEP) pack.

Not legal advice. Rule citations from 46 CFR Part 541 as published on eCFR (current to Oct 5 2026) and 46 U.S.C. 41310; 46 CFR 545.5 summarized, verify full text before production. Charge Complaint procedures updated by FMC final rule published Sept 1 2026.